DDD-T-86b
AUGUST 24, 1959

SUPERSEDING
Int. Fed. Spec. DDD-T-0086a (GSA-FSS)
June 23, 1959 and
Fed. Spec. DDD-T-86
August 26, 1936

FEDERAL SPECIFICATION

TAPE, TEXTILE, COTTON, GENERAL PURPOSE, (GRAY, BLEACHED OR DYED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification governs the requirements for general use cotton tape.

1.2 Classification.

1.2.1 *Type and classes.*—The cotton tape shall be of one type and three classes as specified (see 6.2):

Class 1.—Unbleached.

Class 2.—Bleached.

Class 3.—Dyed.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU-P-268—Paper, Kraft, Wrapping.

CCC-T-191—Textile Test Methods.

PPP-B-591—Boxes, Fiberboard, Wood-Cleated.

PPP-B-601—Boxes, Wood, Cleated, Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Boxes, Fiber.

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the

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Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–B–10377—Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes and Appendix.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Other publications. — The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Munsell Book of Color.

(Application for copies should be addressed to the Munsell Color Company, 10 East Franklin Street, Baltimore, Maryland.)

3. REQUIREMENTS

3.1 Samples.

3.1.1 *Standard sample.*—Unless otherwise specified (see 6.5), the tape shall be equal to the standard sample in all respects not specifically covered by this specification.

3.1.2 *Preproduction sample.*

3.1.2.1 *Civil agencies.*—When specified, a preproduction sample shall be submitted for approval.

3.1.2.2 *Military agencies.* — Unless otherwise specified, before production is commenced, the contractor shall submit for approval to the contracting officer, a 5-linear-yard sample of tape he proposes to furnish (see 6.2).

3.2 Material.

3.2.1 *Cotton.*—The cotton shall be of suitable length of staple and grade to meet the requirements of this specification.

3.2.2 *Yarns.*—The cotton used in the manufacture of the yarns shall be thoroughly cleaned, well carded or combed, evenly drawn and spun into single or plied yarn for both the warp and filling.

3.2.3 *Yarn sizes.*—The yarn sizes for tape shall be singles 20, or equivalent, for the warp, and singles 30, or equivalent, for the filling when tested as specified in 4.4.1. A tolerance of plus or minus 5 percent will be allowed for the specified yarn size.

3.3 Color.—The tape shall be unbleached, bleached or dyed as specified (see 6.2). Unless otherwise specified, the color shall be obtained by the use of vat dyes. Sulfur dyes or dyes containing sulfur compounds capable of oxidation to sulfuric acid are prohibited (see 4.5.1).

3.3.1 *Dyeing.*—The colored tape shall be stock, yarn, or piece dyed. When olive drab No. 7 is specified, the tape shall be dyed to match an approved shade standard (see 6.2 and 6.3).

3.3.2 *Matching.*—The color shall match the approved standard shade under natural

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(north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2800° Kelvin.

3.3.3 *Colorfastness.*

3.3.3.1 *Rating with standard shade sample.*—Unless otherwise specified (see 6.2), the dyed tape shall show colorfastness to laundering, chlorine bleaching and light after 40 standard fading hours and crocking equal to, or better than, the standard sample (see 6.2) when tested as specified in table III.

3.3.3.2 *Rating without standard sample.*—When no standard sample is available (see 6.2), the dyed tape shall show good colorfastness to laundering, chlorine bleaching and light after 40 standard fading hours and no more crocking than Munsell value 8.5 when tested as specified in table III. When black is specified, the tape shall show no more crocking than Munsell value 6.5 when tested for wet crocking.

3.4 Physical requirements.—The physical requirements of the tape shall conform to the requirements as specified in table I and as otherwise specified herein. The width of the tape shall be as specified (see 6.2). A plus or minus tolerance of 1/32 inch for widths up to 1 inch and a plus 1/16 or minus 1/32 inch for widths of 1 inch and wider is allowed on the widths of the tapes in table I.

3.4.1 *Weave.*—The weave shall be a two up and two down single or multiple herringbone twill with one or more reversals of the twill across the width of the tape. The tapes shall be woven with a selvage on both sides.

3.5 Nonfibrous material.—The total sizing, finishing, and other nonfibrous material shall not exceed 8.5 percent for class 1 tape, and 2.5 percent for class 2 tape, and 3.5 percent for class 3 tape when tested as specified in table III.

TABLE I.—*Physical requirements*

Width	Total ends, minimum	Filling yarns per inch, minimum	Warp strength, minimum
Inches			*Pounds*
1/4	28	38	25
3/8	36	38	30
7/16	44	38	35
1/2	52	38	42
5/8	60	38	50
11/16	68	38	58
3/4	76	38	65
13/16	84	38	72
7/8	92	38	80
1	100	38	85
1 1/16	108	38	90
1 1/8	116	38	95
1 1/4	135	38	115

3.6 Put-up. — Unless otherwise specified, the tape shall be evenly wound on standard commercial spools containing a minimum of 720 yards of tape 1 inch or less in width. For tape in widths greater than 1 inch the length shall be a minimum of 504 yards per spool. A plus tolerance of 4 percent in length is allowed. Each spool shall contain no more than three pieces of tape and no piece shall be less than 50 yards in length.

3.7 Workmanship.—The tape shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable quality levels established by this specification.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General. — Sampling, inspection, and test procedures for components or materials described shall be in accordance with this specification and with subsidiary specifications referenced herein to the extent applicable, except that this specification shall govern in the event of conflict. Sampling,

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inspection, and test procedures contained in this section are applicable to inspection by and for the Government.

4.2 Contractor's inspection responsibility. —Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and made available to the Government as specified in the contract or order.

4.3 Preacceptance inspection.

4.3.1 *Preproduction sample inspection.* — The preproduction sample submitted in accordance with 3.1.2, shall be visually inspected for appearance, color and finish. The sample shall be tested for physical and chemical characteristics specified in table III in accordance with the referenced test methods.

4.4 Inspection for acceptance. — Except when otherwise specified by civil Government agencies, inspection shall be in accordance with the provisions of Military Standard MIL–STD–105 except where otherwise indicated.

4.4.1 *Testing of components.*—In addition to the quality assurance provisions of the subsidiary specifications, the warp and filling yarns (3.2.3) shall be tested for yarn number in accordance with method 4021 of Federal Specification CCC–T–191. The unit of product for testing shall be of one cone, tube, or spool each of the warp and the filling yarns. The lot size shall be expressed in units of one cone, tube, or spool each. Except when otherwise specified by civil Government agencies, the sample size shall be in accordance with Inspection Level L1, table VII of Appendix to Military Standard MIL–STD–105. The acceptance quality level shall be 6.5 test failures per 100 units. When specified in the procurement document, a contractor's certificate of compliance may be accepted by the Government as evidence of conformance with the yarn sizes as specified in 3.2.3.

TABLE II.

Lot size in spools	Sample size in spools	Maximum number of defects acceptable in sample
5 and under	1	0
6 to 15	2	0
16 to 40	3	0
41 to 110	5	0
111 to 300	7	0
301 and over	10	1

4.4.2 *Inspection of the end item.*

4.4.2.1 *Examination of the end item.*—Defects found during the examination shall be classified in accordance with 4.4.2.1.1, 4.4.2.1.2 and 4.4.2.1.3.

4.4.2.1.1 *Yard-by-yard examination.*—The required yardage of each piece shall be inspected on both sides and visual defects classified as listed herein. All defects found shall be counted regardless of their proximity one to another, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be one linear yard. Except when otherwise specified by civil Government agencies, the sample size shall be in accordance with Inspection Level II, table III, of Military Standard MIL–STD–105. The acceptable quality level shall be 0.4 major defects and 1.5 total (major and minor combined) defects per 100 units (yards). The lot size shall be expressed in units of one yard each. An approximate equal number of yards shall be examined from each roll selected.

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Classification of defects

Glossary No. (Fed. Std. No. 4)	Defects	Classification	
		Major	Minor
79	Abrasion marks—resulting in a weak place	X	
33	Broken or missing end:		
	Multiple, 1 inch or more in length	X	
	Single, 1 inch or more in length		X
	Multiple, more than ½ inch but not more than 1 inch in length	X	
	Single, 1 inch or more in length		X
11	Broken or missing pick:		
	Two or more contiguous, regardless of width	X	
	Single, more than ½ the width of the tape		X
90, 91, 92	Cut, hole, or tear—any	X	
59, 61	Crease or wrinkle, permanently embedded		X
4	Double pick—extending over full width of tape		X
69, 70	Dye stain or streak—clearly visible*		X
31, 32	Fine thread or drop ply—less than ½ the normal thickness		X
13, 34	Float or skips:		
	Multiple, more than ¼ inch in combined warp and filling directions	X	
	Multiple, ¼ inch or less in combined warp and filling directions		X
	Single, floating over 1 inch or more in length		X
	Single, over more than ½ the width of the tape		X
2	Heavy filling bar—clearly visible		X
3, 5, 27	Jerked-in or sloughed-off filling, slub or slug—more than twice the thickness of normal yarn		X
6, 28	Kink or loop, on surface of tape		X
7, 29	Knots:		
	Large, improperly pieced knots, clearly visible*		X
	Protruding from surface or with untrimmed tails extending from surface of tape		X
9	Light or fine filling bar—clearly visible*	X	
10	Mispick:		
	Two or more within ½ inch	X	
	Single		X
55, 56	Selvage defects:		
	Missing on either edge	X	
	Cut, frayed or torn edges, any	X	
49, 54	Loopy or loose edge:		
	Continuous loops extending over 4½ inches or more, resulting in loose ends or ends within edge	X	

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Classification of defects—(Cont'd)

Glossary No. (Fed. Std. No. 4)	Defects	Classification	
		Major	Minor
49, 54 (Cont'd)	Loopy or loose edge:		
	Any intermittent or contiguous loops extending over 1 inch or more in length, except continuous loops extending over more than 4½ inches of the edge		X
	Regular or irregular floats due to skips in weave on outermost selvage edge, continually repeating for 4½ inches or more	X	
34	Floats of outermost selvage edge, continually repeating for more than 1 inch but less than 4½ inches		X
	Outermost ends not firmly held to body of tape, easily slipping along the length when pulled with needle	X	
	Outermost ends not firmly held to body of tape but not slipping from position when pulled with needle		X
52	Rolled or wavy edge which will not permanently flatten under manual pressure		X
51	Scalloped edge—clearly visible*		X
38	Slack end—extending over more than 9 inches in length		X
39	Slack warp—causing distortion, bagginess of tape or irregular positioning of filling	X	
78	Smash—any	X	
74, 75, 76	Spot, stain or streak, clearly visible*		X
15, 37	Tight end or tight pick:		
	Causing warping, doubling or rolling	X	
	Clearly visible* but not causing warping or rolling		X
	Width:		
	More than 1/16 inch narrower than specified	X	
	More than 1/32 inch but not more than 1/16 inch narrower than specified		X
	More than 1/32 inch wider than specified		X
40	Wrong draw:		
	Two or more ends ½ inch or more in length	X	
	Single end, 9 inches or more in length		X

* At normal inspection distance (approximately 3 feet).

4.4.2.1.2 *Overall examination.*—Each defect listed below shall be counted no more than once in each spool examined. The unit of product for this examination shall be one spool. The sample size and acceptance number shall be as shown in table II.

Defects

Off shade, not within established tolerances.
Uneven dyeing.
Poor penetration.
Shaded throughout.
Objectionable odor.
Overall uncleanness.

4.4.2.1.3 *Length examination.*

4.4.2.1.3.1 *Examination for length of individual spools.*—Each defect listed herein shall be counted no more than once in each spool

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examined. The unit of product for this examination shall be one spool. The sample size and acceptance number shall be as shown in table II.

Defects

Less than 720 yards per spool for tape 1 inch or less, in width.
More than 749 yards per spool for tape 1 inch or less, in width.
Less than 504 yards per spool for tape more than 1 inch in width.
More than 524 yards per spool for tape more than 1 inch in width.
More than 3 pieces per spool.
Any piece less than 50 yards in length.
Actual gross length more than 2 yards less than gross length marked on spool ticket.

4.4.2.1.3.2 *Examination for total yardage in sample.*—The lot shall be unacceptable if the total of the actual gross lengths of spools in the sample is more than 5 yards less than the total of the gross lengths marked on spool tickets.

4.4.3 *Testing of the end item.*—The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table III and 4.5.1 shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The acceptable quality level for each characteristic shall be 6.5 test failures per 100 units. Except when otherwise specified by civil Government agencies the sample size shall be in accordance with Inspection Level L–1, table VII of the Appendix to Military Standard MIL–STD–105. The lot size shall be expressed in units of 1 linear yard. The unit of product for test purposes shall be 5 linear yards.

4.5 Additional test methods.

4.5.1 *Presence of sulfur dyes.*—Sulfur dyes are determined by tests for available sulfur.

4.5.1.1 To preclude interference in dye identification, sample first shall be boiled in distilled water to remove any trace of chemical used in the dyeing process.

4.5.1.2 Dissolve 100 grams of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water. One to 1.5 grams of the finished tape, cut into small pieces, shall be used for the test. Place the sample in a test tube of suitable size and cover with the stannous chloride solution. Place a filter paper which has been moistened with one drop of lead acetate solution (made by dissolving 5 grams of reagent grade lead acetate to 100 cc. of distilled water; add a drop of glacial acetic acid to clear up if cloudy), over the mouth of the test tube (punch needle hole in paper). Heat the test tube slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur.

5. PREPARATION FOR DELIVERY

For civil agency procurement Federal Standard 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

5.1 Packaging.—Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 *Level A.*—Each spool of tape shall be wrapped with a band of transparent plastic film or with minimum 30-pound kraft paper conforming to grade B of Federal Specification UU–P–268. The wrap shall completely inclose the tape on the spool and shall be secured with gummed or pressure-sensitive tape.

5.1.2 *Level B.*—Unless otherwise specified, the requirements of 5.1.1 shall be used.

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TABLE III.—*Test requirement*

Characteristic	Specification references		Number determinations per unit of product	Results reported as:
	Requirement	Test method		
Weave	3.4.1	Visual	1	Pass or fail.
Thread count:				
Total ends	table I	5050	1	Actual count.
Picks per inch	table I	5050	5	Average of five determinations to nearest whole number.
Breaking strength	table I	5100 [1]	5	Average of five determinations to nearest 1.0 pounds.
Nonfibrous material	3.5	2611	2	Average of two determinations to nearest 0.1 percent.
Dye identification	3.3	[2]		[2]
Use of dye formulation (OD 7 only)	6.3	[2]		[2]
Presence of sulfur	3.3	4.5.1	1	Pass or fail.
Colorfastness to:				
Laundering	3.3.3	5610	1	Standard sample—satisfactory or unsatisfactory. No standard—good, fair or poor.
Chlorine bleach	3.3.3	5600	1	Standard sample—satisfactory or unsatisfactory. No standard—good, fair or poor.
Light (40 S.F.H.)	3.3.3	5660	1	Standard sample—satisfactory or unsatisfactory. No standard—good, fair or poor.
Crocking	3.3.3	5651 [4]	1	Standard sample—satisfactory or unsatisfactory. No standard—value nearest to Munsell values standard.
Crocking	3.3.3	5650 A [3]		Satisfactory or unsatisfactory.

[1] Breaking strength shall be determined on the full width of the tape and the jaw dimensions shall be greater than the width of the tape.

[2] A contractor's certificate of compliance is applicable for this characteristic (see 6.2).

[3] For civil agencies.

[4] Method B (good) delete value 3.5 and substitute value 8.5.

5.1.3 *Level C.*—Tape shall be packaged in accordance with good commercial practice.

5.2 Packing.—Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.*—Tape, of one class and width only, packaged as specified in 5.1, shall be packed in snug-fitting fiberboard shipping containers constructed, closed, and strapped to conform to class 2, V3c or V3s, style RSC of Federal Specification PPP–B–636. Alternatively, shipping containers may be used that are constructed and strapped to conform to overseas type of Federal Specification PPP–B–591, PPP–B–601, or Military Specification

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MIL–B–10377; or class 2, style 2 or 4 of Federal Specification PPP–B–621. Each shipping container shall be provided with a type I or II, grade B or C, class 2 sealed case liner conforming to Military Specification MIL–L–10547. The net weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.2 *Level B.*—Tape, of one class and width only, packaged as specified in 5.1, shall be packed in snug-fitting fiberboard shipping containers constructed to conform to type I or II, class 1, style RSC of Federal Specification PPP–B–636, except that the minimum bursting strength of the fiberboard shall be 275 p.s.i. All flaps of the fiberboard shipping containers shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive. Alternatively, shipping containers may be used that are constructed to conform to domestic type of Federal Specification PPP–B–591, PPP–B–601 or Military Specification MIL–B–10377; or class 1, style 2 or 4 of Federal Specification PPP–B–621. The net weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.3 *Level C.*—Tape, packaged as specified in 5.1, shall be packed in a manner acceptable to common carriers for safe transportation to point of destination specified in shipping instructions at the lowest transportation rate for such supplies. Shipping container shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 **Marking.**—In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Federal Standard No. 123.

5.3.1 *Military requirements.*—In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Military Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.**—The tape covered by this specification is utilized in the fabrication of clothing and other textile items.

6.2 **Ordering data.** — Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Class required (see 1.2).
(*c*) Color required (see 3.3).
(*d*) When preproduction sample is not required (see 3.1.2.2).
(*e*) When certification of yarn size is required (see 4.4.1).
(*f*) When certification of vat dyes is required (see table III).
(*g*) Colorfastness required if other than specified (see 3.3.3).
(*h*) When standard sample is not available (see 3.3.3.2).
(*i*) Width of tape required (see 3.4).
(*j*) Put-up if other than specified (see 3.6).
(*k*) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.3 A suggested, but not mandatory, dye stuff formulation for Olive Drab No. 7 is as follows:

Vat Olive T.
Vat Olive Green B, Pr. No. 293 or Vat Olive GGL.
Vat Khaki 2G, Pr. No. 122.

Shaded with either or both of the following:
Vat Brown R. C. I. No. 1151 or Vat Brown BR, Pr. No. 118.
Vat Yellow of suitable fastness.

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6.4 In the preparation of contracts or orders, it should be noted that various options, choices and alternatives, as indicated in Federal Specification PPP–B–636, PPP–B–591 and PPP–B–601 may be exercised in the preparation for delivery of this commodity.

6.5 For access to the standard shade sample address the procuring office issuing the invitation for bids.

6.6 **Transportation description.** — The transportation description applicable to this item is:

Rail:

Tape, cotton, not otherwise indexed by name.

Carload minimum weight 20,000 pounds subject to rule 34 Uniform Freight Classification.

Motor:

Tape, cotton.

Motor volume minimum weight—Any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsover; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—E M O Q Sig
Navy—S MC Sh
Air Force: WADC

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